

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihalan Produk: **Palladium on calcium carbonate, poisoned with 3.5% lead**  
 Product Description: **Palladium on calcium carbonate, poisoned with 3.5% lead**  
 Cat No. : 195070000; 195070100; 195070500  
 Synonyms: Lindlar Catalyst.  
 CAS No 7440-05-3  
 Molecular Formula Pd

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

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 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                      |                      |
|------------------------------------------------------|----------------------|
| Acute oral toxicity                                  | Category 4 (H302)    |
| Acute Inhalation Toxicity - Dusts and Mists          | Category 4 (H332)    |
| Reproductive Toxicity                                | Category 1A (H360Df) |
| Specific target organ toxicity - (repeated exposure) | Category 2 (H373)    |
| Chronic aquatic toxicity                             | Category 2 (H411)    |

**Label Elements**


Signal Word

Danger

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## Hazard Statements

H360Df - May damage the unborn child. Suspected of damaging fertility  
H373 - May cause damage to organs through prolonged or repeated exposure  
H411 - Toxic to aquatic life with long lasting effects  
H302 + H332 - Harmful if swallowed or if inhaled

## Precautionary Statements

### Prevention

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

### Response

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P308 + P313 - IF exposed or concerned: Get medical advice/attention

### Storage

P405 - Store locked up

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component         | CAS No    | Weight % |
|-------------------|-----------|----------|
| Calcium carbonate | 471-34-1  | 91.5     |
| Palladium         | 7440-05-3 | 5        |
| Lead              | 7439-92-1 | 3.5      |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

#### Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

#### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

#### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

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## Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

## Indication of any immediate medical attention and special treatment needed

**Notes to Physician** Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical.

### Extinguishing media which must not be used for safety reasons

No information available.

### Special hazards arising from the substance or mixture

Explosive properties.

### **Hazardous Combustion Products**

Lead, lead oxides.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

### Environmental precautions

Do not flush into surface water or sanitary sewer system. Should not be released into the environment. Do not allow material to contaminate ground water system.

### Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Avoid dust formation. Use only under a chemical fume hood. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

### Conditions for Safe Storage, Including any Incompatibilities

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Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Protect from direct sunlight.

## Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component | Malaysia | ACGIH TLV                   | OSHA PEL                  |
|-----------|----------|-----------------------------|---------------------------|
| Lead      |          | TWA: 0.05 mg/m <sup>3</sup> | TWA: 50 µg/m <sup>3</sup> |

| Component | European Union                   | The United Kingdom                                                      | Germany                                                                                |
|-----------|----------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Lead      | TWA: 0.15 mg/m <sup>3</sup> (8h) | STEL: 0.45 mg/m <sup>3</sup> 15 min<br>TWA: 0.15 mg/m <sup>3</sup> 8 hr | TWA: 0.004 mg/m <sup>3</sup> (8 Stunden).<br>MAK<br>Höhepunkt: 0.032 mg/m <sup>3</sup> |

### Exposure Controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles)

#### Hand Protection

Protective gloves

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Respiratory Protection

No protective equipment is needed under normal use conditions

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

### Environmental exposure controls

Prevent product from entering drains Do not allow material to contaminate ground water system Local authorities should be advised if significant spillages cannot be contained

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

#### Appearance

Dark grey

#### Physical State

Powder Solid

#### Odor

Odorless

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|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Odor Threshold                          | No data available        |                                   |
| pH                                      | No information available |                                   |
| Melting Point/Range                     | No data available        |                                   |
| Softening Point                         | No data available        |                                   |
| Boiling Point/Range                     | No information available |                                   |
| Flash Point                             | No information available | Method - No information available |
| Evaporation Rate                        | Not applicable           | Solid                             |
| Flammability (solid,gas)                | No information available |                                   |
| Explosion Limits                        | No data available        |                                   |
| Vapor Pressure                          | No information available |                                   |
| Vapor Density                           | Not applicable           | Solid                             |
| Specific Gravity / Density              | No data available        |                                   |
| Bulk Density                            | No data available        |                                   |
| Water Solubility                        | Insoluble                |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Autoignition Temperature                | Not applicable           |                                   |
| Decomposition Temperature               | > 600°C                  |                                   |
| Viscosity                               | Not applicable           | Solid                             |
| Explosive Properties                    | No information available |                                   |
| Oxidizing Properties                    | No information available |                                   |
| Molecular Formula                       | Pd                       |                                   |
| Molecular Weight                        | 106.42                   |                                   |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

|                          |                                          |
|--------------------------|------------------------------------------|
| Hazardous Polymerization | Hazardous polymerization does not occur. |
| Hazardous Reactions      | None under normal processing.            |

### Conditions to Avoid

Incompatible products.

### Incompatible Materials

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Acids. Alcohols.

## Hazardous Decomposition Products

Lead. lead oxides.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

Oral

Category 4

Dermal

Based on available data, the classification criteria are not met

Inhalation

Category 4

### Toxicology data for the components

| Component         | LD50 Oral                 | LD50 Dermal                                | LC50 Inhalation                           |
|-------------------|---------------------------|--------------------------------------------|-------------------------------------------|
| Calcium carbonate | LD50 = 6450 mg/kg ( Rat ) | > 2000 mg/kg (Rat)<br>(OECD Guideline 402) | > 3 mg/L 4h (Rat)<br>(OECD Guideline 403) |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

##### (d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

| Component                              | Test method                                       | Test species | Study result    |
|----------------------------------------|---------------------------------------------------|--------------|-----------------|
| Calcium carbonate<br>471-34-1 ( 91.5 ) | OECD Test Guideline 429<br>Local Lymph Node Assay | mouse        | non-sensitising |

No information available

(e) germ cell mutagenicity; No data available

| Component                              | Test method                                                | Test species | Study result |
|----------------------------------------|------------------------------------------------------------|--------------|--------------|
| Calcium carbonate<br>471-34-1 ( 91.5 ) | Bacterial Reverse Mutation Test<br>OECD Test Guideline 471 | in vitro     | negative     |

(f) carcinogenicity; No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component | EU | UK | Germany | IARC     |
|-----------|----|----|---------|----------|
| Lead      |    |    |         | Group 2A |

(g) reproductive toxicity;  
Teratogenicity

Category 1A

Teratogenic effects have occurred in experimental animals.

(h) STOT-single exposure; No data available

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(i) STOT-repeated exposure; Category 2

Target Organs Central nervous system (CNS), Blood, Kidney.

(j) aspiration hazard; Not applicable  
Solid

Other Adverse Effects The toxicological properties have not been fully investigated.

Symptoms / effects, both acute and delayed No information available.

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### Ecotoxicity effects

The product contains following substances which are hazardous for the environment. Contains a substance which is: Very toxic to aquatic organisms. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component         | Freshwater Fish                                                                                                                                                          | Water Flea                         | Freshwater Algae | Microtox |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|------------------|----------|
| Calcium carbonate | LC50 > 56 mg/L/96h                                                                                                                                                       |                                    |                  |          |
| Lead              | LC50: = 1.32 mg/L, 96h static (Oncorhynchus mykiss)<br>LC50: = 1.17 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: = 0.44 mg/L, 96h semi-static (Cyprinus carpio) | EC50: = 600 µg/L, 48h (water flea) |                  |          |

### Persistence and degradability

Persistence  
Degradability  
Degradation in sewage treatment plant

Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary  
Insoluble in water, May persist.  
Not relevant for inorganic substances.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

### Bioaccumulative potential

May have some potential to bioaccumulate; Product has a high potential to bioconcentrate

### Mobility in soil

Spillage unlikely to penetrate soil. Is not likely mobile in the environment due its low water solubility.

### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

Other adverse effects No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

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## Waste treatment methods

### Waste from Residues/Unused Products

Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

### Other Information

Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains Do not let this chemical enter the environment

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN3077  
Hazard Class 9  
Packing Group III  
Proper Shipping Name ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. Palladium on calcium carbonate, poisoned with 3.5% lead

### Road and Rail Transport

UN-No UN3077  
Hazard Class 9  
Packing Group III  
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### IATA

UN-No UN3077  
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### Special Precautions for User

No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

X = listed

| Component         | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|-------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Calcium carbonate | 207-439-9 | X    | X   | X     | X    | X    | X     | X    | KE-04487 |
| Palladium         | 231-115-6 | X    | X   | X     | X    |      | X     | X    | KE-27744 |
| Lead              | 231-100-4 | X    | X   | X     | X    |      | X     | X    | KE-21887 |

| Component | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Lead      |                                                                                           |                                                                                          |                            | Annex I - Y31                      |

### National Regulations

ACR19507



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**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Revision Date**

21-Mar-2025

**Revision Summary**

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### **Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**